

AUNG MINN KHANT

AI Engineer — Computer Vision & Medical Imaging | M.Sc. Data Science (AI Applications)

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SUMMARY

M.Sc. Data Science (AI Applications) student at the University of Macau with a B.Sc. in Electrical and Computer Engineering. Hands-on experience designing and training 3D U-Net and nnU-Net v2 segmentation models for PET/CT and SPECT/CT scans on multi-GPU clusters (NVIDIA A100, RTX A6000), with rigorous evaluation workflows (five-fold cross-validation, Dice, HD95) and Docker-based reproducible training.

TECHNICAL SKILLS

ML & AI: PyTorch, TensorFlow, nnU-Net, MONAI, scikit-learn

Programming Languages: Python, JavaScript, C, MATLAB

DevOps & Systems: Docker, Linux, Git, CI/CD, GPU Clusters (A100, RTX A6000), VMware, Virtualization

Data & Tools: SQL, NIFTI, ITK-SNAP

Web Development: React, Node.js, REST APIs, MongoDB

EXPERIENCE

System Administrator / Research Assistant — Biomedical Imaging Laboratory, University of Macau Sep 2024 – Present

- Administer production GPU servers (NVIDIA A100 80GB, RTX A6000) supporting concurrent ML training for multiple researchers
- Monitor GPU utilization with nvidia-smi to identify workloads and resolve resource contention issues
- Apply minimally invasive maintenance practices to avoid disrupting active training jobs during business hours
- Maintain organized port-mapping documentation and a container registry to prevent conflicts and enable quick user onboarding
- Maintain and update laboratory websites and provide day-to-day IT support for the research team

Web Development Instructor — Lat Twaee Education 2020 – 2021

- Designed and delivered a web development curriculum to 30+ students; 90% completed capstone projects
- Created course materials covering JavaScript, React, and full-stack development fundamentals
- Mentored students through debugging, code reviews, and project development

Freelance Web Developer — ShalYaung Digital 2019 – 2022

- Delivered custom web solutions for SMEs across multiple industries
- Translated client requirements into technical specifications and functional applications
- Managed the end-to-end project lifecycle from requirements gathering to deployment

PROJECTS

Deep Learning Image Segmentation Pipeline — Final-Year Project

Python, PyTorch, nnU-Net, Docker, NVIDIA A100 / RTX A6000

- Designed and trained 3D U-Net and nnU-Net v2 models for volumetric image segmentation on PET/CT and SPECT/CT scans
- Trained for 1,000 epochs across 5+ public and private datasets; completed five-fold cross-validation and evaluated each fold on its held-out test set
- Tracked Dice and HD95 metrics and reviewed best- and worst-case segmentations in ITK-SNAP to diagnose failure modes
- Implemented a Docker-based training workflow on a multi-GPU cluster with patch-based sampling (128³) and data augmentation
- Ran persistent training sessions with MONAI, CUDA, and tmux; monitored Dice, IoU, precision, recall, and HD95

Full-Stack Web Applications — Freelance & Projects

React, Node.js, JavaScript, REST APIs, MongoDB

- Developed 5+ responsive websites for SMEs using React and Node.js, improving page load times by 40%
- Built RESTful APIs and integrated third-party services for e-commerce and business applications
- Implemented responsive UI/UX designs with modern frontend optimization techniques

EDUCATION

M.Sc. Data Science (AI Applications) 2026 – 2028 (Expected)

University of Macau

B.Sc. Electrical and Computer Engineering 2022 – 2026

University of Macau

CERTIFICATIONS & LANGUAGES

Certifications: CS50x: Introduction to Computer Science — Harvard University, 2020

Languages: English (Fluent), Mandarin (Intermediate), Burmese (Native)